Cpts 317. 10/10

Key Lecture.

Last time: (Pumping Lemma) Let L be an inf. reg language. Then there is a number n s.t. for any word z in L with $|z| \geq n$, there is a way to write the z into

$$z = uvw$$

s.t. $|uv| \leq n$, $|v| \geq 1$ and for all $i \geq 0$, we have

$$uv^iw \in L.$$

Show: $L = \{a^n b^n c^n : n \geq 0\}$ is not reg.

Proof. (use PL). Assume that L is reg. Let x be the number in the PL. Pick $z = a^n b^n c^n \in L$.

Assume that $z = uvw$ with $|uv| \leq n$, $|v| \geq 1$. Then,

$$z = \overbrace{a \cdots a}^{\leftarrow x \rightarrow} \overbrace{b \cdots b}^{\leftarrow x \rightarrow} \overbrace{c \cdots c}^{\leftarrow x \rightarrow}$$

$$z = \underbrace{uv}_{\leq n} w$$

you can also try: $L = \{a^n b^n : n \geq 0\}$ is not reg.

the z picked must dep. on x, must be in L,

We can see that the entire uv is contained in the a^n block in the z.

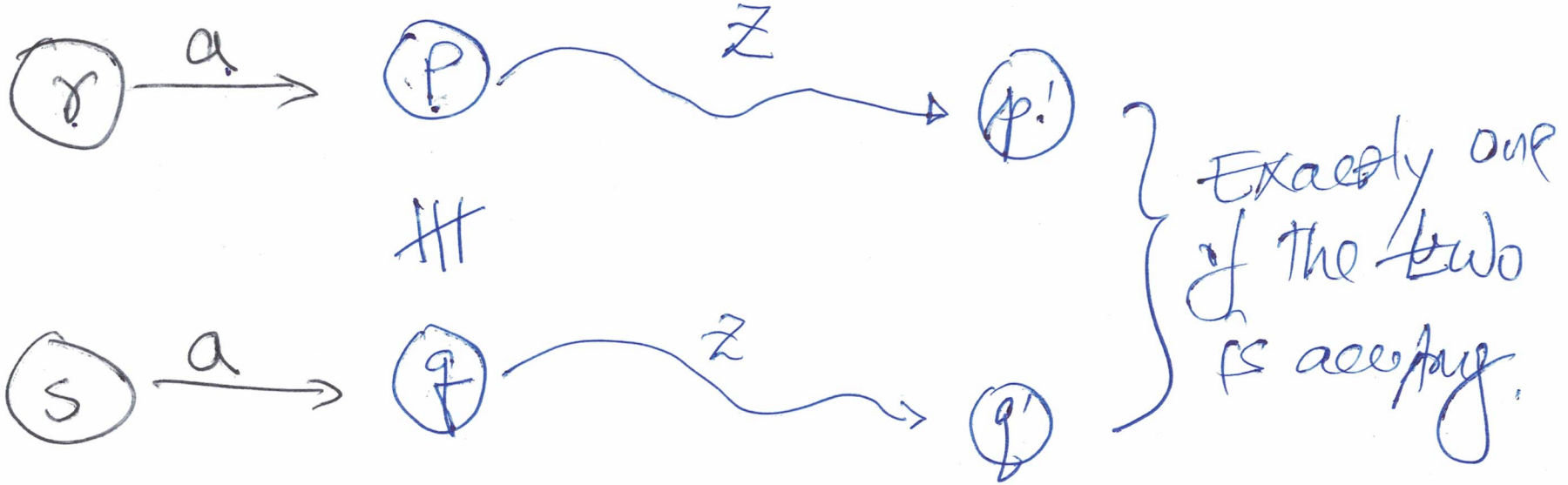
That is, uv contains a's only. Then, we can see $uvvw \notin L$ since $uvvw$ is $a^m b^n c^n$ with $m = n + |v| > n$. A contradiction to PL.

$\boxtimes$

Minimizing a FA.

Given a DFA M, to find a DFA M' s.t. $L(M) = L(M')$ and M' has minimal # of states.

Intuition: need to find states p, q s.t. $p \neq q$ and $p \equiv q$. (So, I can merge p with q) (to reduce the # of states).



Def: $p \not\equiv q$ when there is a word z such, starting p after reading z, M enters a state p', also, starting from q after reading z, M enters state q', and exactly one of p' and q' is accepting.

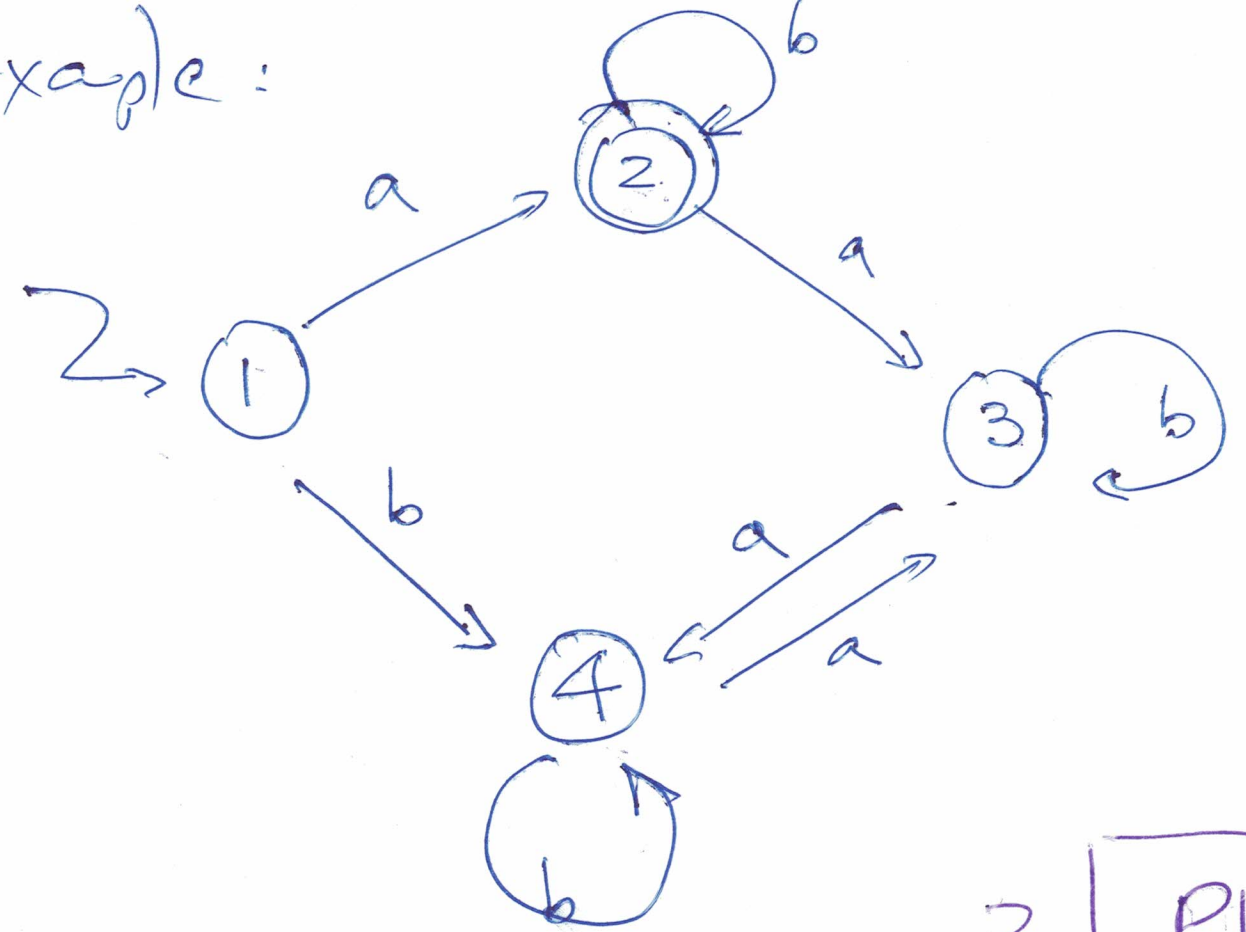
Assume that $p \not\equiv q$. and

$$\delta(r, a) = p,$$
$$\delta(s, a) = q.$$

then, $r \not\equiv s$.

Base case: $p \not\equiv q$ when ~~p~~ p is accepting, q is not accepting.

Exaple:

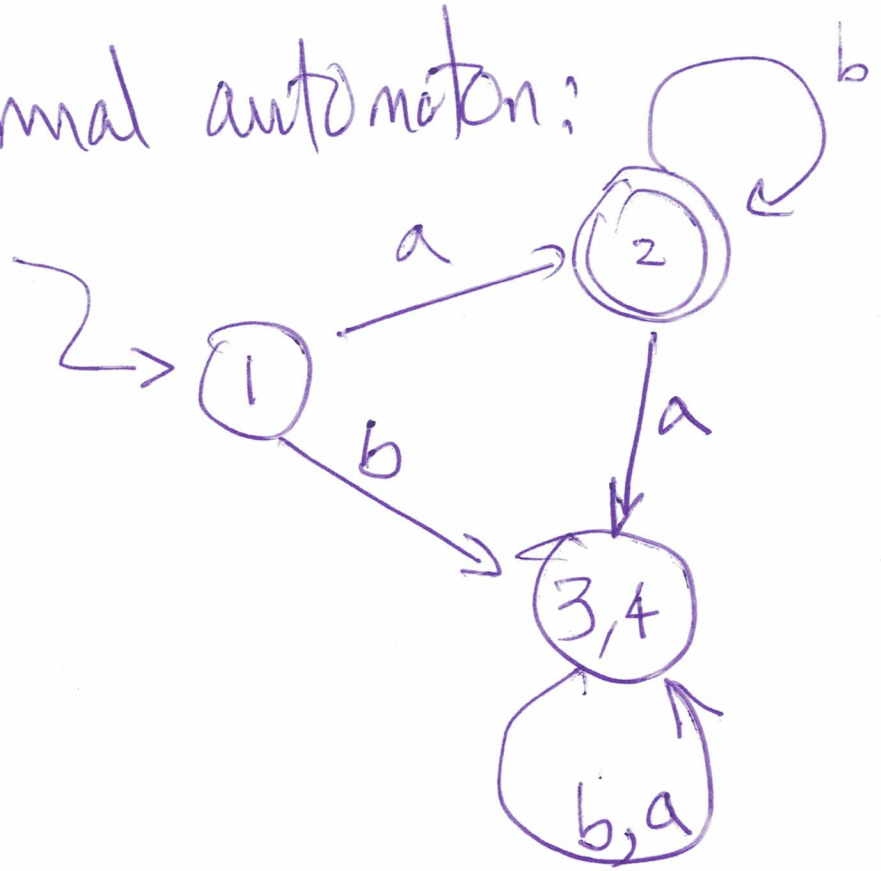


P1: base cases.

2	P1		
3	P2	P1	
4	P2	P1	
	1	2	3

so, state 3 $\equiv$ state 4.

minimal automaton:



Summary:

$$\text{NFA} \longrightarrow \text{DFA} \longrightarrow \text{minimal DFA}.$$

exp. blowup.

(we are not good at understanding nondet.)

efficient.

Summary:

Reg. languges are closed under all Boolean operations: $+$, $\cap$, $\overline{}$

complement.

Algorithms:

Exples:

(1). Show an efficient alg to decide:

Given: a FA M

Qstion: $L(M) = \phi$?

Input to the alg ? M

output of the alg ? . yes/No .